EXPERIMENT - 2

AIM: Convert the ER diagram into a relational schema. Write complete MySQL. CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample

SOURCE CODE:

```
-- 1. CUSTOMER
CREATE TABLE Customer (
    customer_id INT AUTO_INCREMENT PRIMARY KEY,
    customer_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE
);

-- 2. MULTIVALUED CUSTOMER PHONE NUMBERS
CREATE TABLE CustomerPhone (
    customer_id INT NOT NULL,
    phone_no VARCHAR(15) NOT NULL,
    PRIMARY KEY (customer_id, phone_no),
    FOREIGN KEY (customer_id)
        REFERENCES Customer(customer_id)
        ON DELETE CASCADE
);

-- 3. ADDRESS
CREATE TABLE Address (
    address_id INT AUTO_INCREMENT PRIMARY KEY,
    customer_id INT NOT NULL,
    house_no VARCHAR(20),
    street VARCHAR(100) NOT NULL,
    city VARCHAR(50) NOT NULL,
    state VARCHAR(50) NOT NULL,
    pin_code VARCHAR(10) NOT NULL,
    FOREIGN KEY (customer_id)
        REFERENCES Customer(customer_id)
        ON DELETE CASCADE
);

-- 4. CATEGORY
CREATE TABLE Category (
```

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```
    category_id INT AUTO_INCREMENT PRIMARY KEY,
    category_name VARCHAR(100) NOT NULL UNIQUE
);

-- 5. SELLER
CREATE TABLE Seller (
    seller_id INT AUTO_INCREMENT PRIMARY KEY,
    seller_name VARCHAR(100) NOT NULL,
    seller_email VARCHAR(100) NOT NULL UNIQUE,
    phone VARCHAR(15) NOT NULL
);

-- 6. PRODUCT
CREATE TABLE Product (
    product_id INT AUTO_INCREMENT PRIMARY KEY,
    product_name VARCHAR(100) NOT NULL,
    price DECIMAL(10,2) NOT NULL,
    description VARCHAR(500),
    stock INT NOT NULL DEFAULT 0,
    category_id INT,
    seller_id INT,

    CONSTRAINT chk_product_price CHECK (price >= 0),
    CONSTRAINT chk_product_stock CHECK (stock >= 0),

    FOREIGN KEY (category_id)
        REFERENCES Category(category_id)
        ON DELETE SET NULL,

    FOREIGN KEY (seller_id)
        REFERENCES Seller(seller_id)
        ON DELETE SET NULL
);

-- 7. PRODUCT SUBTYPE: ELECTRONICS
CREATE TABLE Electronics (
    product_id INT PRIMARY KEY,
    brand VARCHAR(100) NOT NULL,
    warranty VARCHAR(50),
    FOREIGN KEY (product_id)
        REFERENCES Product(product_id)
        ON DELETE CASCADE
);
```

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```sql
-- 8. PRODUCT SUBTYPE: CLOTHING
CREATE TABLE Clothing (
    product_id INT PRIMARY KEY,
    size VARCHAR(20) NOT NULL,
    fabric VARCHAR(50) NOT NULL,
    FOREIGN KEY (product_id)
        REFERENCES Product(product_id)
        ON DELETE CASCADE
);

-- 9. PRODUCT SUBTYPE: GROCERY
CREATE TABLE Grocery (
    product_id INT PRIMARY KEY,
    expiry_date DATE NOT NULL,
    FOREIGN KEY (product_id)
        REFERENCES Product(product_id)
        ON DELETE CASCADE
);

-- 10. PRODUCT SUBTYPE: BOOKS
CREATE TABLE Books (
    product_id INT PRIMARY KEY,
    author VARCHAR(100) NOT NULL,
    isbn VARCHAR(20) NOT NULL UNIQUE,
    FOREIGN KEY (product_id)
        REFERENCES Product(product_id)
        ON DELETE CASCADE
);

-- 11. ORDERS
CREATE TABLE Orders (
    order_id INT AUTO_INCREMENT PRIMARY KEY,
    customer_id INT NOT NULL,
    order_date DATE NOT NULL,
    order_status VARCHAR(30) NOT NULL DEFAULT 'Pending',
    total_amount DECIMAL(10,2) NOT NULL,

    CONSTRAINT chk_order_amount CHECK (total_amount >= 0),

    FOREIGN KEY (customer_id)
        REFERENCES Customer(customer_id)
        ON DELETE CASCADE
```

```
);

-- 12. ORDER ITEM (WEAK ENTITY)
CREATE TABLE OrderItem (
    order_id INT NOT NULL,
    product_id INT NOT NULL,
    quantity INT NOT NULL,
    unit_price DECIMAL(10,2) NOT NULL,

    PRIMARY KEY (order_id, product_id),

    CONSTRAINT chk_item_quantity CHECK (quantity > 0),
    CONSTRAINT chk_item_price CHECK (unit_price >= 0),

    FOREIGN KEY (order_id)
        REFERENCES Orders(order_id)
        ON DELETE CASCADE,

    FOREIGN KEY (product_id)
        REFERENCES Product(product_id)
        ON DELETE CASCADE
);

-- 13. PAYMENT
CREATE TABLE Payment (
    payment_id INT AUTO_INCREMENT PRIMARY KEY,
    order_id INT NOT NULL UNIQUE,
    payment_date DATE,
    amount DECIMAL(10,2) NOT NULL,
    payment_method VARCHAR(30) NOT NULL,
    payment_status VARCHAR(30) NOT NULL DEFAULT 'Pending',

    CONSTRAINT chk_payment_amount CHECK (amount >= 0),

    FOREIGN KEY (order_id)
        REFERENCES Orders(order_id)
        ON DELETE CASCADE
);

-- 14. DELIVERY
CREATE TABLE Delivery (
    delivery_id INT AUTO_INCREMENT PRIMARY KEY,
```

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```
    order_id INT NOT NULL UNIQUE,
    delivery_date DATE,
    delivery_status VARCHAR(30) NOT NULL DEFAULT 'Pending',
    tracking_no VARCHAR(50) UNIQUE,

    FOREIGN KEY (order_id)
        REFERENCES Orders(order_id)
        ON DELETE CASCADE
);



-- CUSTOMER
INSERT INTO Customer (customer_name, email) VALUES
('Umair Rahman', 'umair@example.com'),
('Rahul Sharma', 'rahul@example.com'),
('Priya Verma', 'priya@example.com');

-- CUSTOMER PHONE NUMBERS
INSERT INTO CustomerPhone VALUES
(1, '9876543210'),
(1, '9123456780'),
(2, '9988776655'),
(3, '9876501234');

-- ADDRESS
INSERT INTO Address
(customer_id, house_no, street, city, state, pin_code)
VALUES
(1, '12A', 'Sector 15', 'Chandigarh', 'Punjab', '160015'),
(1, '22B', 'Sector 34', 'Chandigarh', 'Chandigarh', '160022'),
(2, '45', 'MG Road', 'Delhi', 'Delhi', '110001'),
(3, '18C', 'Hazratganj', 'Lucknow', 'Uttar Pradesh', '226001');

-- CATEGORY
INSERT INTO Category (category_name) VALUES
('Electronics'),
('Clothing'),
('Grocery'),
('Books');

-- SELLER
INSERT INTO Seller (seller_name, seller_email, phone) VALUES
```

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```
('Tech World', 'tech@example.com', '9811111111'),
('Fashion Hub', 'fashion@example.com', '9822222222'),
('Daily Mart', 'daily@example.com', '9833333333'),
('Book House', 'books@example.com', '9844444444');

-- PRODUCT
INSERT INTO Product
(product_name, price, description, stock, category_id, seller_id)
VALUES
('Laptop', 55000.00, 'Laptop with 16GB RAM', 10, 1, 1),
('T-Shirt', 799.00, 'Cotton T-shirt', 50, 2, 2),
('Rice Bag', 1200.00, '10 kg rice bag', 30, 3, 3),
('DBMS Book', 650.00, 'Database textbook', 20, 4, 4);

-- PRODUCT SUBTYPES
INSERT INTO Electronics VALUES (1, 'Dell', '2 Years');
INSERT INTO Clothing VALUES (2, 'L', 'Cotton');
INSERT INTO Grocery VALUES (3, '2027-12-31');
INSERT INTO Books VALUES (4, 'Korth', '9780133970777');

-- ORDERS
INSERT INTO Orders
(customer_id, order_date, order_status, total_amount)
VALUES
(1, '2026-10-01', 'Delivered', 55799.00),
(2, '2026-10-02', 'Pending', 1200.00),
(1, '2026-10-03', 'Shipped', 650.00);

-- ORDER ITEMS
INSERT INTO OrderItem VALUES
(1, 1, 1, 55000.00),
(1, 2, 1, 799.00),
(2, 3, 1, 1200.00),
(3, 4, 1, 650.00);

-- PAYMENTS
INSERT INTO Payment
(order_id, payment_date, amount, payment_method, payment_status)
VALUES
(1, '2026-10-01', 55799.00, 'UPI', 'Paid'),
(2, NULL, 1200.00, 'COD', 'Pending'),
(3, '2026-10-03', 650.00, 'Card', 'Paid');
```

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```sql
-- DELIVERIES
INSERT INTO Delivery
(order_id, delivery_date, delivery_status, tracking_no)
VALUES
(1, '2026-10-04', 'Delivered', 'TRK1001'),
(2, NULL, 'Pending', 'TRK1002'),
(3, NULL, 'Shipped', 'TRK1003');



-- Display all customers
SELECT * FROM Customer;

-- Display customers and their addresses
SELECT c.customer_name, a.city, a.state, a.pin_code
FROM Customer c
JOIN Address a ON c.customer_id = a.customer_id;

-- Display products with category and seller
SELECT p.product_name, p.price,
       c.category_name, s.seller_name
FROM Product p
LEFT JOIN Category c ON p.category_id = c.category_id
LEFT JOIN Seller s ON p.seller_id = s.seller_id;

-- Display orders with customer details
SELECT o.order_id, c.customer_name,
       o.order_date, o.total_amount, o.order_status
FROM Orders o
JOIN Customer c ON o.customer_id = c.customer_id;

-- Calculate order totals from order items
SELECT order_id,
       SUM(quantity * unit_price) AS calculated_total
FROM OrderItem
GROUP BY order_id;

-- Display order, product, and payment details
SELECT o.order_id, p.product_name,
       oi.quantity, oi.unit_price,
       pay.payment_method, pay.payment_status
FROM Orders o
```

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```
JOIN OrderItem oi ON o.order_id = oi.order_id
JOIN Product p ON oi.product_id = p.product_id
LEFT JOIN Payment pay ON o.order_id = pay.order_id;



-- TEST 1: SET NULL
-- Deleting a category preserves its products.
DELETE FROM Category WHERE category_id = 1;

SELECT product_id, product_name, category_id
FROM Product;

-- The Laptop remains, but category_id becomes NULL.



-- TEST 2: CASCADE
-- Deleting a customer deletes their addresses,
-- phone numbers, and orders.
DELETE FROM Customer WHERE customer_id = 2;

SELECT * FROM Address;
SELECT * FROM CustomerPhone;
SELECT * FROM Orders;

-- Rahul's dependent records are deleted.

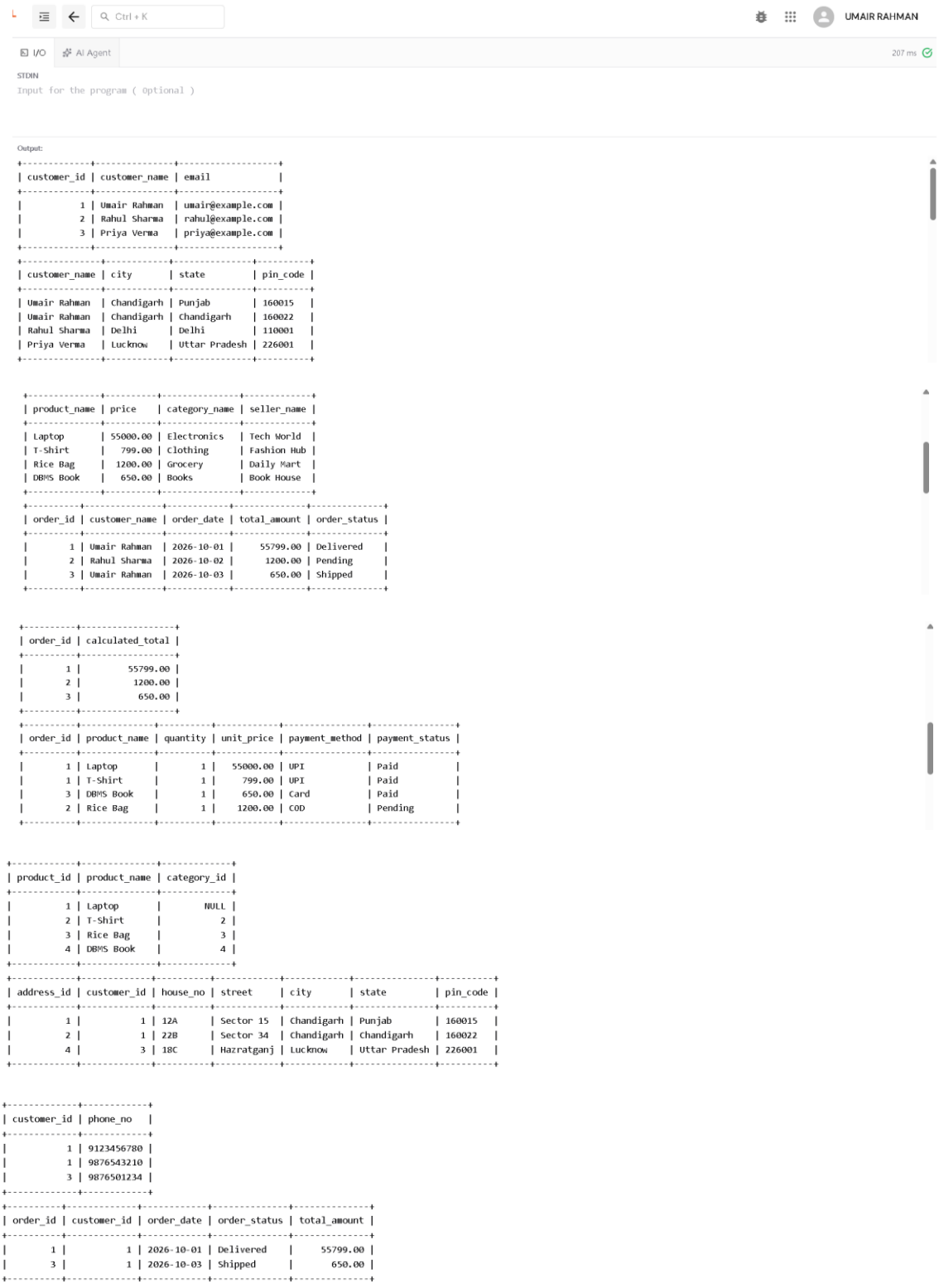


-- TEST 3: Product subtype cascade
-- Deleting a product also deletes its subtype.
DELETE FROM Product WHERE product_id = 2;

SELECT * FROM Clothing;
```

OUTPUT:

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Ctrl + K

UMAIR RAHMAN

I/O | AI Agent | 207 ms

STDIN

Input for the program (Optional)

Output:

```
+-------------+---------------+--------------------+
| customer_id | customer_name | email              |
+-------------+---------------+--------------------+
|           1 | Umair Rahman  | umair@example.com  |
|           2 | Rahul Sharma  | rahul@example.com  |
|           3 | Priya Verma   | priya@example.com  |
+-------------+---------------+--------------------+
+---------------+------------+---------------+----------+
| customer_name | city       | state         | pin_code |
+---------------+------------+---------------+----------+
| Umair Rahman  | Chandigarh | Punjab        | 160015   |
| Umair Rahman  | Chandigarh | Chandigarh    | 160022   |
| Rahul Sharma  | Delhi      | Delhi         | 110001   |
| Priya Verma   | Lucknow    | Uttar Pradesh | 226001   |
+---------------+------------+---------------+----------+
```

```
+--------------+----------+---------------+-------------+
| product_name | price    | category_name | seller_name |
+--------------+----------+---------------+-------------+
| Laptop       | 55000.00 | Electronics   | Tech World  |
| T-Shirt      |   799.00 | Clothing      | Fashion Hub |
| Rice Bag     |  1200.00 | Grocery       | Daily Mart  |
| DBMS Book    |   650.00 | Books         | Book House  |
+--------------+----------+---------------+-------------+
+----------+---------------+------------+--------------+--------------+
| order_id | customer_name | order_date | total_amount | order_status |
+----------+---------------+------------+--------------+--------------+
|        1 | Umair Rahman  | 2026-10-01 |     55799.00 | Delivered    |
|        2 | Rahul Sharma  | 2026-10-02 |      1200.00 | Pending      |
|        3 | Umair Rahman  | 2026-10-03 |       650.00 | Shipped      |
+----------+---------------+------------+--------------+--------------+
```

```
+----------+------------------+
| order_id | calculated_total |
+----------+------------------+
|        1 |         55799.00 |
|        2 |          1200.00 |
|        3 |           650.00 |
+----------+------------------+
+----------+--------------+----------+------------+----------------+----------------+
| order_id | product_name | quantity | unit_price | payment_method | payment_status |
+----------+--------------+----------+------------+----------------+----------------+
|        1 | Laptop       |        1 |   55000.00 | UPI            | Paid           |
|        1 | T-Shirt      |        1 |     799.00 | UPI            | Paid           |
|        3 | DBMS Book    |        1 |     650.00 | Card           | Paid           |
|        2 | Rice Bag     |        1 |    1200.00 | COD            | Pending        |
+----------+--------------+----------+------------+----------------+----------------+
```

```
+------------+--------------+-------------+
| product_id | product_name | category_id |
+------------+--------------+-------------+
|          1 | Laptop       |        NULL |
|          2 | T-Shirt      |           2 |
|          3 | Rice Bag     |           3 |
|          4 | DBMS Book    |           4 |
+------------+--------------+-------------+
+------------+-------------+----------+-----------+------------+--------------+----------+
| address_id | customer_id | house_no | street    | city       | state        | pin_code |
+------------+-------------+----------+-----------+------------+--------------+----------+
|          1 |           1 | 12A      | Sector 15 | Chandigarh | Punjab       | 160015   |
|          2 |           1 | 22B      | Sector 34 | Chandigarh | Chandigarh   | 160022   |
|          4 |           3 | 18C      | Hazratganj | Lucknow   | Uttar Pradesh | 226001   |
+------------+-------------+----------+-----------+------------+--------------+----------+
```

```
+-------------+------------+
| customer_id | phone_no   |
+-------------+------------+
|           1 | 9123456780 |
|           1 | 9876543210 |
|           3 | 9876501234 |
+-------------+------------+
+----------+-------------+------------+--------------+--------------+
| order_id | customer_id | order_date | order_status | total_amount |
+----------+-------------+------------+--------------+--------------+
|        1 |           1 | 2026-10-01 | Delivered    |     55799.00 |
|        3 |           1 | 2026-10-03 | Shipped      |       650.00 |
+----------+-------------+------------+--------------+--------------+
```

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